

Machine Learning Approaches for Neuroblastoma Risk Prediction and Stratification

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ABSTRACT: Machine learning (ML) holds great promise in advancing risk prediction and stratification for neuroblastoma, a highly heterogeneous pediatric cancer. By utilizing large-scale biological and clinical data, ML models can detect complex patterns that traditional approaches often overlook, enabling more personalized treatments and better patient outcomes. Various ML techniques, such as support vector machines, random forests, and deep learning, have shown superior performance in predicting survival, relapse, and treatment responses in neuroblastoma patients compared to conventional methods. However, challenges like limited data size, model interpretability, data variability, and difficulties in clinical integration hinder broader adoption. Additionally, ethical concerns related to bias and privacy must be addressed. Future work should focus on improving data quality, enhancing model transparency, and conducting thorough clinical validation. With these advancements, ML has the potential to revolutionize neuroblastoma care by refining early diagnosis, risk assessment, and therapeutic decision-making.

KEY WORDS: machine learning, neuroblastoma, risk prediction, personalized treatment, data variability

I. INTRODUCTION

Neuroblastoma (NB) is a prevalent solid tumor in young children, representing around 8% of cancers in pediatric patients, with an estimated 150–200 new cases each year in Japan. This condition is more common in boys, although the genetic and epigenetic factors underlying this sex disparity are still not well understood. The occurrence rate of NB in Japan is comparable to that in the United States, but both the frequency and characteristics of the disease are strongly correlated with race. In North America, NB is less prevalent among black individuals than white individuals, yet those of African descent tend to develop a more aggressive form of the disease than those of European descent. However, precise incidence rates across different ethnic groups are not well established.^{1,2} NB is unique among solid tumors because of its diverse biological features. Genetically, it often shows chromosomal abnormalities, including deletions on chromosomes 1p, 11q, and 14q and gains on chromosome 17q, all of which are linked to a poorer prognosis. NBs may also display specific genetic

alterations, such as *MYCN* amplification and mutations in the *ALK* and *PHOX2B* genes. Other genetic changes include rearrangements in the *ATRX* and *TERT* loci. *MYCN* amplification, along with *ATRX* and *TERT* alterations, is commonly found in high-risk patients. When these alterations are combined with mutations in the *RAS* and/or p53 pathways, the tumor becomes more aggressive, and the prognosis worsens. Additionally, dysregulated gene expression, particularly the overexpression of *MYCN*, *ALK*, *PHOX2B*, and *LIN28B*, plays a crucial role in the initiation and progression of NB by disrupting the balance between cell proliferation and differentiation.^{3,4} Transcriptional and epigenetic studies have further revealed distinct states of cellular differentiation in NB that are heavily influenced by genetic events and epigenetic modifications during tumor development, which reshape the regulatory landscape of NB.⁴ This distinct NB biology also affects clinical outcomes, ranging from spontaneous regression to aggressive, treatment-resistant disease. Patients are classified into low-, intermediate-, and high-risk groups on the basis of historical survival rates and specific histological

and biological criteria. Low-risk patients typically have excellent survival with minimal treatment. Intermediate-risk patients, who are primarily treated with surgery and chemotherapy, have improved outcomes, with ongoing efforts to reduce therapy on the basis of biological markers. High-risk patients, representing approximately half of new cases, require aggressive multimodal treatment, including chemotherapy, surgery, radiotherapy, and immunotherapy. Despite these interventions, the 4-year event-free survival rate is approximately 59%, highlighting the need for further research into the origins of NB and new treatments.^{3,5}

As a result, machine learning (ML) has become a powerful tool for analyzing complex genetic and clinical data, helping to identify patterns that enhance outcome prediction, guide treatment strategies, and personalize patient care.⁶ While many prognostic models for NB have been developed using various gene expression profiles to assess disease risk and prognosis, we still cannot predict 100% of cases. For example, Oberthuer and team used support vector machines (SVMs) to classify high-risk NB patients via microarray data from 709 NB samples.⁷ The model was built using 75 tumors with different characteristics and outcomes and then validated on the remaining 634 samples via Kaplan–Meier and multivariate Cox regression analyses. The classifier achieved an accuracy of 0.95 and was particularly effective for low- and intermediate-risk patients. Nevertheless, because of the complex molecular mechanism associated with NB,^{8,9} identifying pathogenic and protective genes from large datasets is challenging.¹⁰ In this review, the authors have summarized the recent applications of ML in NB research. It starts by outlining the pathogenesis associated with NB and emerging ML techniques to predict and classify NB. Specifically, it covers how ML methods are used to determine the tumor's origin, classify molecular subtypes, and predict patient outcomes. Finally, this study addresses the current challenges and suggests strategies for integrating ML into routine clinical practice for improved NB management.

II. DEVELOPMENT AND PATHOGENESIS OF NB

NBs originate from neural crest cells, a group of multipotent stem cells.¹¹ During embryonic

development, neural crest cells migrate and differentiate into various types of cells, including melanocytes, sensory neurons, enteric neurons, and sympathetic neurons. These developing neurons undergo significant programmed cell death (PCD) in the final stages of maturation, with approximately half of them dying through apoptosis. The surviving neurons continue to mature into functional cells, such as ganglion cells. NB exclusively originates from precursor or stem cells within the sympathoadrenal lineage, which is committed to becoming sympathetic neurons. This specificity suggests that the cancerous transformations leading to NB occur after these cells have already committed to differentiating into sympathetic neurons and not from other neural crest-derived lineages.¹

Recent data from single-cell RNA sequencing analyses, which compared malignant NB with neural crest cell signatures, identified two distinct immature cell populations within the neural crest as potential origins of NB.^{12–14} One study revealed that NB predominantly exhibits a chromaffin-cell-like phenotype, suggesting that these cells are the source of the disease.¹⁴ In contrast, two other studies proposed that NBs more closely resemble normal fetal adrenal neuroblasts (also known as sympathoblasts) at the transcriptional level.^{12,13} NB tumors typically arise from chromosomal alterations, which can be either acquired through random cellular events or inherited through genetic predisposition. While most cases of NB are due to acquired mutations, inherited forms, though rare, are linked to specific genetic factors. One prominent marker in high-risk NB is *MYCN* gene amplification, which is found in 20–25% of such tumors and is known for its association with aggressive disease. However, *MYCN* expression is influenced by factors other than amplification. Inherited NB has been connected to mutations in transcription factors such as *PHOX2B* and *ALK*. Interestingly, the overexpression of *TrkA*, a factor involved in apoptosis and angiogenesis, is associated with a more favorable prognosis in pediatric patients. Furthermore, various genetic abnormalities, including chromosomal imbalances and losses or gains on specific chromosome arms, significantly contribute to the development of NB.¹⁵

III. ML APPROACHES

Artificial intelligence (AI) encompasses a range of computational techniques that leverage algorithms to analyze large datasets, enabling tasks such as classification, prediction, and the extraction of valuable insights. Within AI, ML specifically involves developing statistical models that learn from real-world data. These models are trained to make predictions or categorize information, often with initial guidance from humans. As they process more data over time, they become increasingly accurate and effective, adapting and refining their performance without explicit programming.¹⁶ Traditionally, ML includes three main techniques: supervised learning, unsupervised learning, and reinforcement learning and details given in Fig. 1 and Table 1.^{17,18}

A. Supervised Learning

In supervised learning, a statistical model is trained to predict outcomes for new, unseen instances by learning from a labeled dataset. During training, the model receives input data along with the correct labels or target values, which guide its learning process. The model continuously adjusts its internal parameters to minimize the difference between its predictions and the actual labels. This iterative process enables the model to learn patterns and relationships within the data. Once the model is trained, it can be generalized from the patterns it has learned to make predictions on new, unlabeled data. For example, in a medical dataset, the model might be

trained to classify tumors as either benign or malignant by learning from a set of labeled examples. It learns to associate certain features, such as tumor size, shape, and cell structure, with the correct classification. Similarly, a model could be trained to predict future infection rates in different regions by learning from historical data. After training, the model can use these learned patterns to make accurate predictions on new cases, even if it has never encountered them before.^{17,19,20} Common techniques in supervised learning include naive Bayes (NB), linear discriminant analysis (LDA), logistic regression (LR), k-nearest neighbor (KNN), support vector machine (SVM), decision tree (DT), random forest (RF), adaptive boosting (AdaBoost), extreme gradient boosting (XGBoost), stochastic gradient descent (SGD), and rule-based classification.²¹

The naive Bayes classifier is a popular and straightforward supervised learning method.²² It assumes that all features are independent of each other when the classes are labeled. However, this assumption often does not hold true in real-world scenarios, which can sometimes limit the classifier's effectiveness. In fields such as medicine and healthcare, features such as symptoms or risk factors are often interconnected. For example, certain symptoms might frequently occur together when a condition is diagnosed, indicating a natural relationship between them. This interdependence goes against the naive Bayes assumption of feature independence, suggesting that more advanced methods might be needed to accurately capture these complex relationships in the data. However, NB remains a valuable tool,

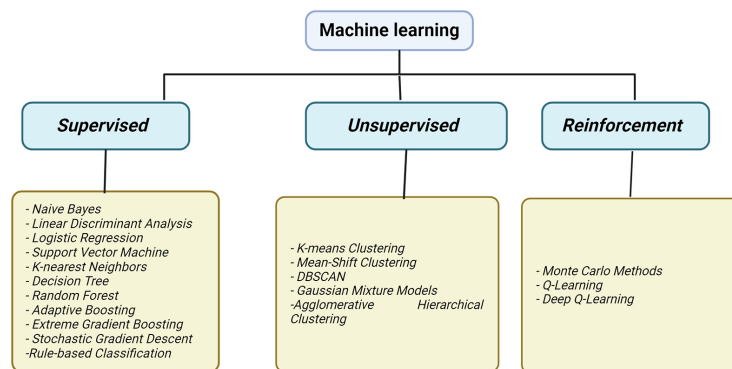


FIG. 1: ML approaches and subtypes

TABLE 1: Different ML approaches

Algorithm	Description	Key characteristics	Advantages	Limitations
Naive Bayes	A probabilistic classifier based on Bayes' theorem with the assumption that features are independent given the class label.	Assumes feature independence; computes class probabilities.	Simple, efficient, and works well with small datasets. Often performs well in text classification.	Assumes features are independent, which is often unrealistic. May perform poorly with correlated features.
Linear discriminant analysis	A dimensionality reduction technique that finds a linear combination of features to maximize class separability.	Projects data to a lower-dimensional space; maximizes class separation.	Effective for feature reduction and visualization. Performs well when classes are linearly separable.	Assumes linear boundaries between classes. May struggle with nonlinearly separable data.
Logistic regression	A model used to estimate the probability of a binary outcome based on one or more predictor variables.	Estimates probabilities using the logistic function; binary classification.	Simple, interpretable, and works well for binary outcomes. Can handle multiple predictors.	Limited to linear relationships; may not perform well with complex, nonlinear data.
Support Vector machine	A classifier that finds the hyperplane that best separates classes in a high-dimensional space.	Constructs a hyperplane to maximize margin between classes; can handle nonlinear data using kernel functions.	Effective for high-dimensional spaces; robust with small sample sizes. Provides good generalization performance.	Computationally intensive for large datasets; requires careful parameter tuning.
k-nearest neighbors	A method that classifies data based on the majority class among its k nearest neighbors in the feature space.	Nonparametric; classification based on nearest neighbors.	Simple and intuitive; effective with small datasets and low-dimensional data.	Computationally expensive for large datasets; sensitive to irrelevant features and choice of k.
Decision tree	A model that splits data into branches based on feature values to make predictions, resulting in a tree-like structure.	Uses a tree structure to model decisions; splits data based on feature values.	Easy to interpret and visualize; handles both numerical and categorical data.	Prone to overfitting, especially with complex trees; may not generalize well to unseen data.
Random forest	An ensemble method that creates a forest of decision trees, each built from a random subset of features, and combines their predictions.	Aggregates predictions from multiple decision trees; uses bagging for training.	Robust to overfitting; handles large datasets well; provides feature importance metrics.	Computationally intensive; less interpretable than individual decision trees.
Adaptive boosting	An ensemble method that combines weak classifiers by focusing on errors of previous models, adjusting their weights.	Combines multiple weak learners into a strong learner; adjusts weights based on errors.	Improves model accuracy; works well with imbalanced datasets.	Sensitive to noisy data; can overfit if not carefully tuned.

TABLE 1: *(continued)*

Extreme gradient boosting	A gradient boosting technique that builds decision trees iteratively to improve accuracy and reduce overfitting.	Uses boosting to iteratively correct errors; optimized for performance and scalability.	High performance and accuracy; handles large datasets effectively; prevents overfitting.	Requires careful tuning of parameters; can be complex to implement and interpret.
Stochastic gradient descent	An optimization algorithm used to minimize the loss function in ML models by updating parameters iteratively.	Updates model parameters based on a randomly selected subset of data (mini-batches).	Efficient for large-scale datasets; reduces training time.	Sensitive to learning rate; may require careful tuning and early stopping to avoid overfitting.
Rule-based classification	A method that creates a set of “if-then” rules to classify data based on feature values.	Uses rule induction to create classification rules; interpretable model.	Provides clear, interpretable rules; easy to understand and implement.	May not capture complex relationships; can become overly complex with many rules.
k-means clustering	A clustering technique that partitions data into k clusters by minimizing the variance within each cluster.	Assigns data points to the nearest cluster center; updates centers iteratively.	Simple and scalable; effective for finding spherical clusters.	Requires specifying the number of clusters; sensitive to initial cluster centers and outliers.
Mean-shift clustering	A nonparametric clustering algorithm that shifts data points toward the peak of the density function to find clusters.	Does not require specifying the number of clusters; identifies clusters based on density.	Flexible; can find clusters of arbitrary shape.	Computationally expensive; performance can be affected by bandwidth parameter choice.
Density-based spatial clustering of applications with noise (DBSCAN)	A clustering method that groups data based on density, identifying clusters of varying shapes and sizes while ignoring noise.	Requires parameters (Eps and MinPts); identifies clusters based on density.	Finds clusters of arbitrary shapes; robust to noise and outliers.	Sensitive to parameter settings; challenging to select optimal Eps value.
Gaussian mixture models (GMMs)	A probabilistic model that represents data as a mixture of multiple Gaussian distributions, estimated using the Expectation-Maximization algorithm.	Models data as a mixture of Gaussian distributions; uses EM algorithm for parameter estimation.	Provides soft clustering; can model complex data distributions.	Assumes Gaussian distribution; can be sensitive to initialization and requires careful tuning.
Agglomerative hierarchical clustering	A clustering technique that starts with individual data points and merges them iteratively based on similarity until all points are in a single cluster.	Builds a hierarchy of clusters; merges clusters based on linkage criteria.	Produces a dendrogram for visualizing cluster relationships; no need to specify the number of clusters in advance.	Computationally intensive for large datasets; validation can be challenging.

TABLE 1: (continued)

Algorithm	Description	Key characteristics	Advantages	Limitations
Monte Carlo methods	Computational techniques that use repeated random sampling to estimate the behavior of complex systems with uncertainty.	Uses simulations to model scenarios and calculate probabilities.	Effective for problems with uncertainty and complex interactions; useful for risk analysis and decision making.	Computationally intensive; results depend on the number of simulations and may have variability.
Q-learning	A reinforcement learning technique that determines the optimal policy by learning the value of state-action pairs.	Updates action-value estimates based on reward feedback to find optimal policy.	Effective for learning optimal policies in multistage decision problems; can handle complex tasks with sequential decisions.	Model misspecification can affect performance; requires extensive exploration and can suffer from large state spaces.
Deep Q-learning	A variant of Q-learning that uses deep neural networks to approximate Q-values, enabling it to handle high-dimensional state spaces.	Combines Q-learning with deep learning to approximate value functions.	Handles complex environments with high-dimensional states; effective in games and other challenging tasks.	Requires substantial computational resources; can be complex to implement and tune.

especially when a quick and simple solution is needed or when the relationships between features are weak or not well understood.

Linear discriminant analysis is used to find a linear transformation that maximizes class separation in a lower-dimensional space. As a supervised dimensionality reduction method, LDA and its variations are widely used to extract key features in various applications, such as human action recognition and biological data classification. However, LDA is not well suited for multilabel problems because of two main challenges. First, accurately determining the contribution of each data point when calculating scatter matrices in multilabel settings is difficult. Second, there can be an imbalance in class sizes, which can distort the separation between classes. To address these issues, multilabel LDA (MLDA) and its adaptations have been developed. These methods incorporate weights that account for the relationships between labels and features, improving the handling of complex multilabel data where multiple labels can be associated with a single instance.²³

Logistic regression is a binary classification technique used for prediction tasks and is based on

a logistic function or a sigmoid function, which is an S-shaped curve. It works by mapping a weighted sum of input features to a value between 0 and 1 via this function. This value represents the probability that a given instance belongs to the positive class. Instead of directly predicting a class label, logistic regression estimates the likelihood of an instance being in a specific class, with the output interpreted as a probability score. For example, a probability of 0.8 indicates an 80% chance that the instance falls into the positive class, providing a nuanced understanding of the classification.²⁴

The support vector machine algorithm, a supervised ML technique, has shown exceptional effectiveness in addressing classification challenges across various biomedical fields, particularly in bioinformatics. Unlike logistic regression, which relies on a preset model to predict the probability of a binary outcome by fitting data to a logistic function, SVM differentiates between two categories by constructing a hyperplane that best separates the classes after the input data are transformed into a high-dimensional space. Given that SVMs operate in a data-driven and model-free manner, they are

highly effective for classification tasks, particularly in scenarios with small sample sizes and many variables. This method has recently been employed to create automated disease classification systems and enhance disease detection techniques in clinical environments.^{25,26}

The k-nearest neighbor algorithm classifies unlabeled data by referencing the features and labels of the training data. It works by identifying the k nearest training data points to a given query and then uses a majority voting method to determine the most common classification among those nearest neighbors. KNN is known for its simplicity and adaptability, making it popular for various classification and regression problems across different data sizes, label counts, noise levels, and contexts. Despite its popularity, the traditional KNN algorithm has several limitations that impact its classification effectiveness. These include its tendency to treat all neighbors equally without considering their relevance, issues with distance calculations between data points, and the inclusion of irrelevant features from the dataset. To address these limitations, various KNN variants have been developed, which enhance the algorithm by optimizing the k parameter, refining distance metrics, assigning weights to different data points, and reducing the dataset size to improve performance.²⁷

The decision tree approach organizes data into a tree-like structure with a root node, internal nodes, and leaf nodes, effectively handling large and complex datasets without the need for a complex parametric framework. In practice, data are often split into training and validation sets. The training set is used to construct the decision tree model, whereas the validation set helps determine the optimal size of the tree for the final model. Decision trees offer a straightforward and intuitive method for analyzing risk factors and calculating sensitivity to outcomes such as mortality. This makes them useful tools for physicians, especially during critical situations such as the COVID-19 pandemic, where they can assist in identifying critically ill patients and making important decisions about ICU resource use and clinical management.^{28,29}

The random forest works by creating a “forest” of decision trees, where each tree is generated

via a process called bagging, and then all trees vote on the class of the data. RF is popular because of its high accuracy, robustness, and ability to rank features by importance. When training an RF model, two important parameters must be set: the number of trees in the forest (ntree) and the number of randomly selected features (mtry) evaluated at each node of a tree. The RF can adjust the voting threshold, which affects the model’s recall, precision, and F score. The model’s performance is measured via the out-of-bag error (OOB), which estimates the misclassification rate on the basis of samples not used during training. One key advantage of RF is its ability to evaluate feature importance, making it useful for understanding which variables are most influential in the classification process.^{30,31}

Another widely recognized algorithm in ML is adaptive boosting, which was introduced by Freund in 1997.³² AdaBoost works by maintaining a set of weights on the training data, which are adjusted after each weak learning cycle, allowing it to adaptively enhance performance. The algorithm generates multiple weak learners on the basis of the idea that combining these weak models can create a more accurate and robust strong learner. Unlike traditional methods such as back-propagation neural networks or convolutional neural networks, AdaBoost-based models can achieve high accuracy, particularly when dealing with multiclass imbalanced datasets.³³ AdaBoost has been widely applied in ensemble learning due to its excellent classification and prediction capabilities, including tasks such as image recognition and modeling complex change predictions.^{33,34}

The extreme gradient boosting algorithm is based on decision trees that work by improving model accuracy through a method called gradient boosting. It builds a series of decision trees where each tree corrects errors made by the previous trees. XGBoost is especially effective because it addresses the problem of overfitting, meaning that it prevents the model from performing well on training data but poorly on new data. The algorithm’s success comes from its special objective function, which optimizes the learning process to increase both accuracy and generalizability.^{35,36}

Stochastic gradient descent has become a widely used method for addressing large-scale supervised ML problems because of its efficiency in reducing training time without compromising accuracy. It is particularly effective when it is applied to standard convex loss functions with regularization terms.³⁷ However, Zhang³⁸ proposed an alternative approach, suggesting that regularization can be handled by early stopping rather than using the traditional regularization term. This method, which is now commonly adopted, involves dividing the training set into smaller subsets—a new training set and a validation set. During each training cycle, the model is tested on the validation set to identify the optimal number of iterations. Once this optimal number is found, the model is trained on the full original dataset for the determined number of iterations, and the performance is evaluated on the test set to assess the method's effectiveness.

Rule induction algorithms are a set of pattern discovery techniques used in supervised ML. In this context, the goal is to uncover reproducible relationships or models from labeled data. Specifically, these algorithms learn how predictor variables (features) are related to one or more target variables that represent discrete classes. Once these relationships are established, the resulting model can predict the class of new observations on the basis of their attribute values. This process of identifying and learning these relationships is known as classification.³⁹ Classification methods can be categorized into two types on the basis of how they represent the learned model: symbolic and subsymbolic approaches.⁴⁰ Symbolic approaches explicitly represent the model via a format that is easily interpretable by humans, such as decision trees and decision rules. In contrast, subsymbolic approaches use structures and formalism that are not directly understandable by humans, such as artificial neural networks, support vector machines (SVMs), or ensemble methods such as random forests (RFs). The choice between symbolic and subsymbolic methods depends on the goal of the analysis: if the primary objective is high classification performance, the method with the best accuracy is typically selected. However, if the goal is to extract and interpret knowledge from the data, methods that provide a more understandable

and interpretable representation of the patterns are preferred.

B. Unsupervised Learning

Unsupervised learning involves working with datasets where a dependent or response variable is not explicitly defined. In this approach, the system analyzes the data to identify patterns without any predefined outcomes. The goal of unsupervised learning is to explore the data structure and generate insights rather than performing statistical hypothesis tests or developing predictive or classification models on the basis of set conditions and responses.⁴¹ Common traditional unsupervised methods include k-means clustering and hierarchical clustering, which organize data into clusters by detecting similarities. Other popular unsupervised techniques used across various fields include k-means clustering, mean-shift clustering, density-based spatial clustering of applications with noise (DBSCAN), Gaussian mixture models (GMMs), and agglomerative hierarchical clustering.²¹

k-means clustering, introduced by Lloyd⁴² and MacQueen,⁴³ is a widely used clustering technique. The algorithm operates as follows: starting with initial, potentially suboptimal clustering, each data point is reassigned to the nearest cluster center. The cluster centers are then updated by calculating the mean of all the points assigned to each center. This process of relocating points and updating centers is repeated until certain convergence criteria are met, such as a predefined number of iterations or minimal changes in the distortion function's value.⁴⁴

The mean-shift algorithm is a nonparametric clustering algorithm that groups data points on the basis of the high-density regions of a data distribution.^{45,46} It works iteratively by shifting data points toward the peak of the density function until convergence. This method does not assume anything about the shape or number of clusters, making it flexible and broadly applicable. However, it can be computationally expensive with a complexity of $O(n^2)$ per iteration, which limits its scalability for large datasets. To improve efficiency, several strategies can be used. One approach is to first cluster a random subset of the data and then assign the remaining data points to these clusters via

methods such as nearest neighbor or discriminant analysis. Another technique is blurring, which smooths out the density function and speeds up convergence by reducing the number of iterations needed. Additionally, density estimates can be computed via random samples from the data. However, this approach may lead to variability and clustering errors because it uses only a portion of the data for density estimation.

DBSCAN⁴⁷ is a popular clustering algorithm known for its ability to find clusters of various shapes and sizes in large datasets, even when there is significant noise.⁴⁸ One of its main advantages is that one does not need to specify the number of clusters beforehand, which makes it flexible for different types of data. DBSCAN requires two key parameters to work: Eps and MinPts. Eps defines the radius of the neighborhood around each data point, whereas MinPts specifies the minimum number of points required within this neighborhood to form a cluster. Although one does not need to know the number of clusters in advance, one does need to set these parameters. Choosing the right Eps value can be challenging, as it significantly affects the clustering results. If Eps is too large or too small, the algorithm might not perform well, especially with datasets that have clusters of varying densities. Therefore, it may be necessary to adjust Eps and MinPts separately for different parts of the dataset to obtain accurate clustering results.

The Gaussian mixture model is a sophisticated technique for analyzing nonGaussian data and is classified as a linear, nonGaussian multivariate statistical method.^{49,50} It operates by modeling the data as a mixture of multiple Gaussian distributions, where each component represents a different cluster or pattern within the data. The number of Gaussian components and their parameters are estimated via the expectation-maximization (EM) algorithm. This algorithm involves two main steps: the expectation step (E-step), which calculates the posterior probabilities on the basis of current parameters and prior probabilities, and the maximization step (M-step), which updates the parameters to maximize the likelihood. This iterative process continues until convergence. Additionally, the GMM can be evaluated via the Bayesian information criterion (BIC), which helps in selecting the optimal number of Gaussian components. For a system with multiple steady states, a

GMM is constructed from normal operating data, and the number of components is estimated iteratively. A contribution index is then calculated for each Gaussian component, and the most significant faulty variables are identified by combining these local indices into a global index through Bayesian inference.

Agglomerative hierarchical clustering is used to understand the structure of data by grouping similar observations. It starts by calculating the dissimilarity between pairs of data points and then combines the closest pairs into clusters until all the points are grouped. The way clusters are merged depends on the chosen linkage method.^{51,52} This method is often used in unsupervised learning to find natural groupings within a dataset, such as identifying patient subgroups. However, validating the results can be tricky, particularly in biomedical applications, as it often relies on examining the clusters after they have been formed rather than on predefined accuracy measures.

C. Reinforcement Learning

Reinforcement learning, which is distinct from supervised and unsupervised learning, mimics human learning through interaction with its environment. The model refines its actions on the basis of continuous feedback received during both the training and execution phases. Initially, actions are exploratory and random, but as the model gathers more feedback, it shifts toward deterministic actions, a process known as exploitation. This method involves balancing exploration (discovering new strategies) and exploitation (using known strategies for immediate rewards). Reinforcement learning is particularly effective in environments such as intensive care units, where it can dynamically adjust treatments on the basis of patient responses. For example, optimal vasopressor dosing is determined through iterative adjustments rather than predefined targets.⁵³ Common reinforcement learning algorithms include Monte Carlo methods, Q-learning, and deep Q-learning.²¹

The Monte Carlo method uses repeated random sampling to estimate the behavior of complex systems with inherent uncertainty.^{54–56} This involves performing many simulations to model different scenarios and calculate probabilities or outcomes on the basis of these simulations. This approach is particularly useful

for problems with intricate interactions, uncertainties, or incomplete information, where traditional analytical methods may be insufficient. Applications include clinical laboratory data analysis, clinical strategy development, and drug development, such as simulating clinical trials. The method helps in understanding the impact of variability and making informed decisions by analyzing a wide range of possible outcomes.

Q-learning is a reinforcement learning technique used to determine optimal treatment strategies in clinical settings by analyzing multistage randomized trials or observational studies.^{57,58} It works by starting at the final stage of the study and determining the best treatment option to achieve the desired outcome. With this optimal treatment fixed, Q-learning then works backward through previous stages, optimizing future treatment choices on the basis of the assumption that all subsequent treatments will also be optimal. This backward induction helps avoid short-term solutions that might lead to worse long-term results. However, the effectiveness of Q-learning can

be compromised by model misspecification, especially as the number of stages increases, potentially resulting in residual confounding and less effective treatment strategies. Q-learning is also commonly applied in decision-making, gambling, and random event handling. It has also been effectively combined with deep learning (DL) and reinforcement learning (RL) to address complex tasks, such as the AlphaGo system for the board game “Go.” Deep Q-networks, a variant of Q-learning, utilize prior human experiences to train and predict value functions, enhancing performance through this learned knowledge.⁵⁹

IV. ML APPROACHES IN NB

Numerous studies have employed ML and statistical approaches to create multigene models or other predictive frameworks aimed at forecasting patient subgrouping, risk, outcomes (such as survival, relapse, and recurrence), and response to treatment (Table 2). It is essential to understand that patient

TABLE 2: Applications of ML in NB research

Objective	Methods	Key findings	Clinical implications	Refs.
Enhance neuroblastoma prognosis prediction	cDNA microarray with 5340 genes, probabilistic classifier	Achieved 89% accuracy in predicting 5-year survival rates; identified two subgroups within intermediate-risk category with significant survival differences (36% vs. 89%, $P < 10^{-4}$)	Provides a cost-effective tool for predicting neuroblastoma prognosis and aids in clinical decision-making.	61
Develop a predictive model integrating transcriptome and methylation data	ELMER analysis, ML algorithms	Created INSS stage related signature (ISRS) with high predictive accuracy; differentiated high and low-risk groups with significant survival differences	Enhances diagnostic accuracy and provides insights into disease mechanisms, potentially improving personalized treatment strategies.	62
Investigate cancer stemness features in neuroblastoma	mRNAsi calculation, LASSO regression	Identified four key genes (<i>ERCC6L</i> , <i>DUXAP10</i> , <i>NCAN</i> , <i>DIRAS3</i>) related to stemness; model achieved AUC of 0.918	The stemness-related gene signature could guide individualized treatments and improve prognostic predictions.	60
Identify prognostic markers for neuroblastoma outcome	Random Forests, recursive feature elimination	Time to maximum response and maximum response to first-line treatment were significant predictors; response time > 9 months associated with better survival	Incorporates these markers into clinical practice to enhance prognosis accuracy and management.	63

TABLE 2: (*continued*)

Improve diagnostic accuracy using histological images	Scale Invariant Feature Transform (SIFT), SVM classifier	Achieved high precision and recall; overall accuracy ranged from 76.58% to 81.76%	Enhances histological assessment accuracy and classification of neuroblastoma subtypes.	65
Integrate MIBG scans and ML for treatment prediction	Convolutional Neural Network (CNN)	CNN model (AUC 0.73) combined with clinical data improved prediction accuracy compared to clinical model alone	Potential for integrating imaging and clinical data to guide treatment decisions.	66
Examine role of migrasomes in neuroblastoma	RNA-seq, ML (MigScore)	MigScore model correlated with immune response and drug prediction; identified MS275 as a promising candidate	Highlights migrasome pathways' role in prognosis and potential for personalized treatment.	67
Address variability in pathological classification	Nuclear instance segmentation, feature extraction, logistic regression	Achieved high recall and precision; model accuracy with AUC of 0.946	Improves reliability of neuroblastoma prognosis by reducing classification variability.	68
Compare supervised models for prognosis prediction	k-means clustering, SVM, Naive Bayes, logistic regression, XGBoost	SVM excelled with gene expression data; XGBoost with CNA data; high C-indices and significant log-rank <i>P</i> values	Provides robust classification models for distinguishing neuroblastoma subtypes with different prognoses.	69

outcomes represent the results of clinical interventions, offering insight into their effectiveness. Consequently, survival and mortality rates are considered components of overall patient outcomes.⁶⁰ For example, in a groundbreaking study aimed at improving NB prognosis prediction, researchers developed a cDNA microarray containing 5340 genes sourced from primary NBs. This array was utilized to analyze 136 tumor samples, resulting in the creation of a probabilistic classifier. This classifier achieved a notable 89% accuracy in predicting 5-year survival rates. K–M analysis further revealed that the microarray could effectively differentiate between two subgroups within the intermediate-risk category, with survival rates of 36% and 89% for unfavorable and favorable predictions, respectively ($P < 10e^{-4}$). Consequently, a gene subset chip was developed, maintaining an accuracy of 88%, which provides a cost-effective and highly accurate tool for predicting NB prognosis and could significantly aid in clinical decision-making by enhancing the stratification of patients on the basis of risk levels.⁶¹

A sophisticated study employed enhancer linking by methylation/expression relationship (ELM) analysis to explore epigenetic differences between stage 4S and stage 4 NB. By analyzing enhancer methylation patterns, researchers identified 67 motifs with differential expression. The INSS stage-related signature (ISRS) was then developed via ML algorithms to create a predictive model for NB diagnosis and prognosis. The ISRS not only improved diagnostic accuracy but also offered valuable insights into the molecular mechanisms underlying spontaneous regression and treatment response, highlighting its potential for personalized treatment strategies and a better understanding of disease progression.⁶² Another multicenter retrospective study applied random forests and recursive feature elimination techniques to electronic records from 3034 NB patients. The analysis identified “time to maximum response” and “maximum response to first-line treatment” as significant predictors of patient outcome and relapse. In particular, among high-risk patients, those with a maximum response time > 9

months exhibited significantly better overall survival and relapse-free survival than did those with a response time of < 9 months. These findings underscore the importance of incorporating these prognostic markers into clinical practice to enhance the management and treatment of NB patients by providing more accurate predictions of outcomes and the potential for relapse.⁶³

In another study focusing on DNA methylation, random forests (RFs) were employed to identify intermediate-risk patients within the low-risk category of NB. By analyzing the Human Methylation 450 K dataset from 493 samples, researchers discovered distinct clustering patterns associated with MYCN amplification. Four patient clusters were identified, including MYCN-amplified and non-amplified stages, with RF classification showing high recall and precision for these groups. Notably, enhancer-annotated regions exhibited strong predictive power, whereas promoter regions were less effective. This RF-based model demonstrated high predictive accuracy and highlighted significant associations between MYCN amplification and DNA methylation patterns, offering potential for improved risk stratification and early diagnosis.⁶⁴ ML techniques have also been applied to histological images to improve NB diagnosis. A study used scale invariant feature transform (SIFT) and support vector machine (SVM) classifiers to analyze H&E-stained tissue microarrays and whole sections from 125 NB patients. The method achieved high precision and recall in categorizing NB subtypes, with accuracies ranging from 76.58% to 81.76% across different threshold settings. The overall precision, recall, and F-measure scores were 83.81%, 86.61%, and 85.19%, respectively. This approach demonstrates the potential of computer-aided diagnostics (CAD) to improve diagnostic accuracy by leveraging advanced image analysis techniques for histological assessments.⁶⁵

A study explored the integration of metaiodobenzylguanidine (MIBG) scans with ML to predict the response to induction chemotherapy in high-risk NB patients. Researchers have developed a convolutional neural network (CNN) model that combines MIBG scan data with clinical information such as age, stage, and MYCN amplification. The CNN

model achieved an area under the curve (AUC) of 0.73, whereas the AUC was 0.63 for the clinical model alone. Combining both models improved the prediction accuracy, demonstrating the value of integrating imaging data with clinical features to guide treatment decisions and potentially enhance patient outcomes.⁶⁶ Another recent study focused on migrasomes, which are cellular structures implicated in NB development. By analyzing RNA-seq datasets from public sources and utilizing migrasome-specific markers, researchers developed a ML model called the MigScore. This model, which is based on 10 algorithms and 101 combinations, was validated on two single-cell datasets. The MigScore model enabled the evaluation of immune infiltration and drug response prediction, identifying MS275, a histone deacetylase inhibitor, as a promising therapeutic candidate. This study highlighted the inverse correlation between the MigScore and positive clinical outcomes, emphasizing the importance of migrasome pathways in NB prognosis and the potential for personalized treatment approaches.⁶⁷ To improve the consistency of NB pathological classification, a study utilized 563 H&E whole slides from 107 patients. The approach involved nuclear instance segmentation, feature extraction (including nucleus morphology and intensity), and logistic regression. The segmentation achieved high recall and precision rates of 98.62% and 98.65%, respectively. ML techniques, including feature reduction and bootstrap resampling, were employed to develop a prognostic model incorporating age and nucleus morphology features. This model demonstrated high accuracy, with an area under the curve (AUC) of 0.946 in both the training and testing datasets, enhancing the reliability of NB prognosis and reducing variability in pathological analysis.⁶⁸ A comparative study assessed the robustness of classification models using gene expression and copy number alteration (CNA) data. k-means clustering was initially used to assign classification labels, and data normalization via Z scores was applied to mitigate platform variability. Four algorithms—SVM, naive Bayes, logistic regression, and XGBoost—were evaluated, with SVM performing best on the gene expression data and XGBoost outperforming the CNA data. Validation with internal and external cohorts yielded high

C indices and significant log-rank P values, indicating that these models reliably distinguish between subtypes with different prognoses. This study underscores the effectiveness of supervised learning algorithms in predicting NB prognosis and improving subtype classification.⁶⁹

V. LIMITATIONS AND FUTURE PERSPECTIVES

ML applications in NB risk prediction and stratification hold significant potential but face several key challenges as well. Data limitations, such as small sample sizes, can undermine the robustness and generalizability of models, whereas poor data quality and inconsistencies affect their accuracy. The heterogeneity of NB, with its genetic and molecular diversity, complicates the development of models that work across diverse patient populations. Model-specific issues such as overfitting and challenges in selecting relevant features also limit performance, while the “black box” nature of some models, especially deep learning, makes their predictions difficult for clinicians to interpret and trust. Clinically, the integration of ML models is hampered by a lack of rigorous external validation, difficulties incorporating them into existing workflows, and complex regulatory approval processes. The evolving understanding of NB biology means that models may need frequent updates to stay current with new prognostic markers. Ethical concerns, such as the potential for bias in ML algorithms and privacy issues related to patient data, further complicate the clinical use of these models. Overcoming these challenges through larger, more diverse datasets, improved model interpretability, and robust clinical validation is critical for the successful adoption of ML in NB risk prediction and patient stratification.

VI. CONCLUSIONS

ML has demonstrated significant potential in improving the accuracy of NB risk prediction and stratification. By leveraging complex genomic, epigenomic, and clinical data, ML models can uncover intricate patterns that can lead to more precise patient categorization, thereby optimizing treatment plans and improving outcomes. Various studies

have showcased the effectiveness of ML techniques in predicting survival, relapse, and therapeutic responses, offering a more sophisticated alternative to traditional prognostic tools. However, several challenges persist, including the limited availability of diverse, high-quality datasets, the “black-box” nature of certain ML models, and difficulties in integrating these models into clinical practice. Furthermore, the biological complexity of NB, along with ethical concerns such as data privacy and potential biases, requires ongoing attention. Addressing these limitations through larger, more representative datasets, improving model transparency, and rigorous clinical validation is essential for realizing the full potential of ML in NB care. With continued research and refinement, ML has the potential to transform NB treatment by enabling more accurate risk assessment and personalized therapeutic strategies.

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